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Neutron radiography and tomography of water distribution in the root zone

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Abstract

Water uptake by roots and resulting water redistribution along the soil profile depend on soil hydraulic properties and root distribution, as well as on the physical, chemical, and biological soil–plant interactions that occur in the root zone. The hydraulic properties of the soil in the root zone are difficult to investigate *in situ* at the needed high spatial resolution, and they still present important open questions. For instance, is there more or less water at the root–soil interface compared to the bulk soil? Neutron radiography (2-D) and tomography (3-D) are efficient methods to answer such questions, providing the possibility to image simultaneously water distributions and root structure *in situ* at high spatial resolution. We planted a lupin and a maize in rectangular boxes filled with sandy soils. The plants were grown for 3 weeks at controlled conditions.

Infiltrated water and subsequent water redistribution were imaged for 5 d at regular intervals by means of neutron radiography and tomography. Soil water-content distributions were quantified from the radiographs after correcting for neutron scattering. The radiographs showed that the water content in the root zone was higher than in the bulk soil both during and after infiltration. Similarly, the tomograms showed localized regions of high water content around some locations of the roots, in particular near the tips of the lupin. Local regions of water depletion, which are expected as a consequence of water uptake, were visible along the main root where laterals branched. These results reflect the complexity of soil–plant–water relations, showing the different properties of bulk soil and root zone, as well as the varying moisture gradients along the root system.

Key words: root water uptake / root–soil interactions / rhizosphere / irrigation / noninvasive imaging

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1 Introduction

Plants move a big amount of water from soil to atmosphere and thus control soil moisture dynamics, playing a key role in the global water cycle (*Clothier* and Green, 1997). Water uptake by plants depends on conductivity and spatial distribution of roots, as well as on water availability and soil hydraulic properties. This results in a strong interaction between plants and soils. Additionally, roots actively modify the soil in their vicinity (root zone) and in their close proximity (rhizosphere). Frequent drying-and-wetting cycles at the root–soil interface, root growth, root exudates, and microbial activity modify the physical, chemical, and biological properties of the rhizosphere (*Gregory*, 2006). For instance, it has been reported that the rhizosphere is characterized by larger pores (*Whalley* et al., 2005), lower porosity (*Guidi* et al., 1985), and higher aggregate strength (*Czarnes* et al., 2000) in comparison to the bulk soil. According to *Young* (1995), presence of mucilage within the rhizosheath increases the water-holding capacity of the rhizosphere. An additional cause of increased water-holding capacity of the rhizosphere is soil compaction induced by root growth, as reported by *Bruand* et al. (1996). Soil compaction likely reduces the pore size and therefore increases the capacity to store water under negative water potentials.

Despite the recognition of the specific properties of the root zone and their key role in plant functioning (*Jones* and *Hinsinger*, 2008), their impact on soil moisture dynamics and water uptake is still unclear, and it is currently neglected in most of the models of root uptake (*Gardner*, 1960; *Doussan* et al., 2006; *Roose* and *Fowler*, 2004; *Levin* et al., 2007; *Javaux* et al., 2008; *Schneider* et al., 2009). The lack of knowledge about root–soil interactions is due to the opaque nature of soil and to the high spatial and temporal resolution needed to investigate the properties of the root zone *in situ* and with living plants. Today, recent advances in imaging techniques offer the possibility to measure root distribution and water-content changes *in situ.* For instance, the spatial distribution of roots can be imaged in three dimensions by X-ray tomography (*Heeraman* et al., 1997; *Moran* et al., 2000; *Pierret* et al., 2003; *Gregory* et al., 2004; *Doussan* et al., 2006; *Perret* et al., 2007). Water distributions around roots grown in thin soil slabs were quantitatively imaged in two dimensions by means of light transmission (*Garrigues* et al., 2006). They described a water extraction front moving downwards through a presaturated soil profile. Roots and water-depletion zones were also observed by means of nuclear-magnetic resonance (*MacFall* et al., 1990; *Pohlmeier*

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et al., 2008). Another technique which offers high contrast for roots and water distribution is based on neutron transmission: neutron radiography (*Menon* et al., 2006; *Oswald* et al., 2008; *Moradi* et al., 2009) and neutron tomography (*Nakanishi* et al., 2005; *Tumlinson* et al., 2008). *Nakanishi* et al. (2005) and *Tumlinson* et al. (2008) observed a transition zone at the root–soil interface which retains more water than the surrounding soil, however, just for a single plot at one time. The authors concluded that further studies are needed to clarify the causes, dynamics, and relevance of wet regions around roots. Also, neutron tomography has been recently investigated to visualize the root system of a tomato plant, but without resolving the soil water distribution (*Matsushima* et al., 2008).

Our objective was to investigate the water distribution next to roots of living plants during irrigation and drying periods. According to *Young* (1995), the root zone may have a higher water-holding capacity than the bulk soil. If so, at the same water potential the rhizosphere will be wetter than the bulk soil. It can be speculated that the rhizosphere is wetter than the bulk soil also during transpiration, as long as the gradients in water potential towards roots needed to drive water into roots are smaller than the difference in water-holding capacity between the rhizosphere and the bulk soil. However, if transpiration rate is high and soil conductivity is low, water depletion can occur along the most active parts of the root system. Additionally, we were interested to image locations of water depletion along the root system, in order to obtain information on the active parts of roots.

2 Materials and methods

2.1 Experimental set-up

We made use of neutron radiography and tomography to image water distribution in the root zone. We planted a lupin (*lupinus albus L.*) and a maize (*zea mays L.*) in rectangular containers filled with a sand mixture. Then we used neutron radiography to image in two dimensions the changes in water content in the samples during irrigation and drying periods. Additionally, on the same samples, we used neutron tomography to image the roots and the soil water distribution in three dimensions. To our knowledge, this is the first time that the 2- and 3-D techniques are combined to investigate soil–plant interactions, with the advantage of capturing both, temporal and spatial dynamics of water distribution and root architecture.

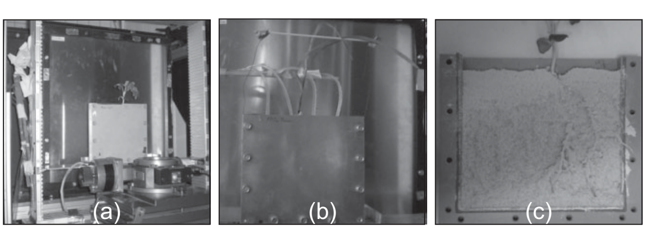

Figure 1: Experimental set-up at Neutra, PSI. (a) Lupin in front of the detector before imaging; (b) maize with tubes for infiltration; (c) opened sample with lupin.

We used three rectangular slabs of 15 cm × 15 cm × 1.5 cm made of aluminum. The rectangular shape was chosen because it allowed imaging of both, fast processes in 2-D and spatial information on root architecture and water distributions in 3-D. Aluminum was chosen because of its low neutron-attenuation coefficient. The small thickness of the *quasi*-2-D design is a requirement for neutron radiography. In fact, thicker samples could not be crossed by thermal neutron, and noise and neutron scattering would increase significantly, making difficult the quantification of water content in the sample (*Hassanein*, 2006). Moreover, 1.5 cm is small enough to justify the two dimensional approximation and large enough to permit root growth (*Moradi* et al., 2009). The sample set-up is shown Fig. 1 a.

The samples were filled horizontally with an artificial mixture of quartz particles of 85% sand, 10% silt, and 5% clay-size fraction. The composition of the soil was a compromise between good growing conditions for the plants, optimal visualization by neutrons, and reduced presence of soil structures such as cracks or dense aggregates. Conventional nutrients were added to the sand mixture for favoring plant growth: NH_4NO_3 (10 mL kg^{-1}), $CaHPO_4$ (350 mg kg^{-1}), K_2SO_4 (5 mL kg^{-1}), $MgCl_2 \cdot 6H_2O$ (5 mL kg^{-1}), $CaSO_4 \cdot 2H_2O$ (420 mg kg^{-1}), and Fe-EDTA (5 mL kg^{-1}).

The water-retention curve of the sand mixture was measured for a similar set-up with the same bulk-soil density by weighing the sample at different water potentials. The water potential was controlled by means of a porous plate placed at the bottom of the sample, and it was varied between 0 and –200 hPa. The conductivity at saturation was calculated by measuring the decreasing height of a water table at the top of the sample, with a potential of –50 hPa being applied at the bottom of the sample. The height of the water table was measured by neutron radiography. For calculating the conductivity, we assumed that the gradient in water potential was constant in time. The decrease of the water table was a few centimeters, which is small compared to the applied gradient of ≈ 70 hPa. This approximation is acceptable for our purposes of estimating the order of magnitude of the conductivity. For a more precise estimation, the falling-head method and its logarithmic solution should be applied. The fitted van Genuchten parameter for the water-retention curve of the sand

Table 1: van Genuchten–fitted parameters for the water-retention curve of the sand mixture, and measured conductivity at saturation bulk-soil density.

Parameter	Value
Residual soil water content θ_{res}	0.05
Saturated soil water content θ_{sat}	0.45
van Genuchten parameter α / cm^{-1}	0.15
van Genuchten parameter n	2.8
Measured saturated hydraulic conductivity K_{sat} / m s^{-1}	3.3×10^{-5}
Measured soil bulk density / cm^3 g^{-1}	1.45
Measured soil bulk density / cm^3 g^{-1}	1.45

mixture, the saturated conductivity, and the soil density are reported in Tab. 1.

One sample was not planted and was used as the control experiment. In one slab we grew a lupin, in another slab we grew a maize. The seeds were germinated for 2 d in a solution of $CaSO_4 \cdot 2H_2O$ on damp paper towels in Petri dishes covered by a dark foil. Then the seeds were placed onto the initial sand mixture in the slabs, and additional sand mixture was added on the top of the samples to cover the seeds.

The plants were illuminated for 16 h d^{-1} by light-emitting diodes (LED). The humidity was 50% and temperature was 22°C. During the growth period, the soil samples were kept moist in order to support plant growth. After 3 weeks, the samples were drip-irrigated by means of four capillary tubes connected to a peristaltic pump (Fig. 1 b). We infiltrated a total amount of 2.60 g of water over 20 min onto the sample with lupin and 2.48 g of water over 19 min onto the sample with maize. These values correspond to an infiltration rate of $\approx 10^{-6}$ m s^{-1} covering the full cross-sectional area. The applied infiltration was thirty times smaller than the saturated conductivity. The average water content in the sample after infiltration was equal to 0.34 m^3 m^{-3} in the sample with lupin and 0.22 m^3 m^{-3} in the sample with maize, as derived from weight measurements.

The water distribution during the 20 min of infiltration and in the following 5 d of root uptake was imaged by means of neutron radiography. Additionally, the samples were tomographed once in order to obtain the three-dimensional root architecture and water distribution.

2.2 Neutron radiography

Neutron radiography is a nondestructive imaging technique with high sensitivity to hydrous materials (*Pleinert* and *Lehmann*, 1997). Hence, neutron radiography is well suited to image distribution of water and roots, which mainly consist of water (*Menon* et al., 2006; *Oswald* et al., 2008; *Moradi* et al., 2009).

The experiments were performed at the neutron-radiography facility NEUTRA, at the Paul Scherrer Institute (PSI), Switzerland. Neutrons are emitted from a lead target which is bombarded with high-energy protons. The produced high-energy neutrons are converted into thermal neutrons by a heavy-water moderator. The neutron flux is between 2.5×10^7 and 3.5×10^6 neutrons cm^{-2} s^{-1}. The beam diameter width can be set from 15 to 40 cm. The beam is parallelized by a collimator and guided to the sample. Neutrons can either be absorbed, scattered, or they can pass undisturbed through the sample. The intensity of the neutron flux behind the sample is detected by a scintillator plate. *Via* a digital camera system, the transmitted neutron flux is converted in gray-value images with values proportional to the neutron flux. These images, called radiograms, carry the information about the thickness of the neutron-attenuating components at each location of the sample, and in first place of water. In order to derive the water content in the sample from the radiograms, several disturbing effects have to be corrected for. The main effect is due to neutron scattering. This has been corrected by means of the QNI software (*Hassanein*, 2006), which is a Monte Carlo–based algorithm that subtracts the effect of neutron scattering. Corrections for other artifacts, such as background scattering or energy-dependent cross-section of the detector, are included in the software tool. For the corrected images, the following exponential law of attenuation is valid:

$$-\log\left(\frac{I(x,z,t)}{I_0(x,z)}\right) = \Sigma_{H_2O}L_{H_2O}(x,z,t) + \Sigma_{Al}L_{Al}(x,z) + \Sigma_{Soil}L_{Soil}(x,z), \quad (1)$$

where x and z are the horizontal and vertical axis perpendicular to the neutron beam, t is time, $I(x,z,t)$ is the intensity of the transmitted beam, $I_0(x,z,t)$ is the intensity of the open beam without the sample, Σ_i and L_i are the neutron-attenuation coefficient and the thickness of the material i, and the subscripts H_2O, Al, and Soil refer, respectively, to water, the aluminum container, and the dry soil. The contribution of $\Sigma_{Soil}L_{Soil}(x,z) + \Sigma_{Al}L_{Al}(x,z)$ is obtained from the radiography of the dry sample. The attenuation coefficient of water is calculated by the QNI software. Then, given the inner thickness of the sample L_{tot} = 1.5 cm, the water content in the sample is calculated as $\theta = L_{H2O} / L_{tot}$ using Eq. 1.

Radiographs were taken at intervals of 1 min during infiltration. After infiltration, radiographies were performed every 3 h for 5 d at 3 a.m./p.m., 6 a.m./p.m., 9 a.m./p.m., and 12/24 noon/midnight. We used a field of view of 279 mm × 279 mm, divided into 1024 × 1024 pixels with pixel side of 0.272 mm. The exposure time for one radiography was equal to 25 s. The samples were weighed before radiography. The weights served as reference to validate the quantification of water content by image analysis.

Additionally, the two samples with plants were tomographed in order to image the water distribution and the root structure in three dimensions. Tomograms were obtained by rotating the sample in steps of 0.5 degrees from 0° to 180° and taking a radiogram at each step. A tomography lasted ≈ 1.5 h. The set of radiograms was reconstructed by means of an IDL (http://www.ittvis.com/)-based tomography-reconstruction package implemented by G. H. Campbell and R. B. Marr. Ring artifacts were removed during the reconstruction process.

The tomography of the sample with maize was performed 3 d after infiltration at 6 a.m. The average water content before tomography was 0.22 m^3 m^{-3}, as derived by weight measurement. The sample with lupin was tomographed 5 d after infiltration at 3 p.m., when the water content was 0.29 m^3 m^{-3}.

3 Results

3.1 Quantification of water content by image analysis

In order to validate the quantification of the soil water content θ by image analysis, we compared the average θ calculated from the radiograms with θ obtained by weighing the sam-

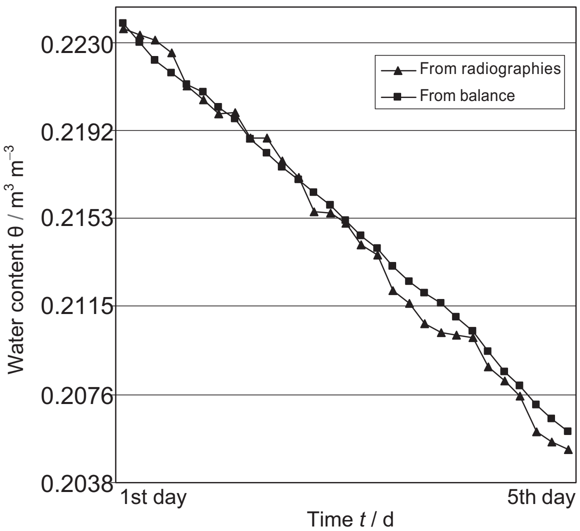

Figure 2: Average water content in the sample with maize during the drying period, as measured by weighing and by neutron radiography.

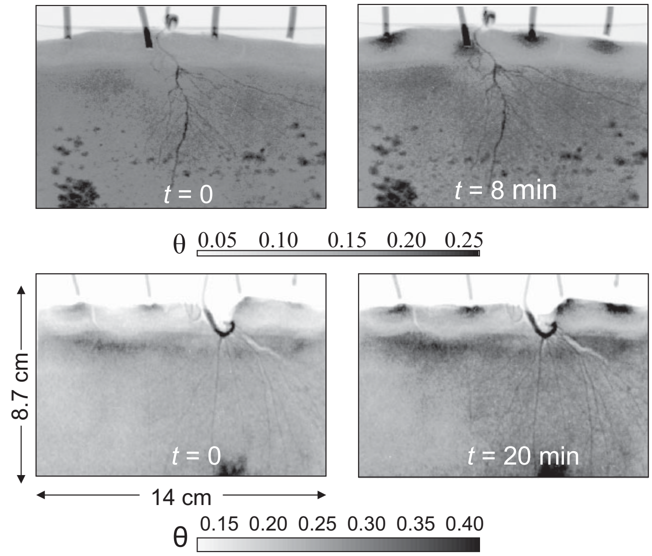

Figure 3: Time-series radiography of water infiltration into sandy soil planted with maize and lupin. Top: sample with maize at the beginning (left) and at the end of infiltration (right). Dark spots in the lower part were caused by precipitates at the container walls. Bottom: sample with lupin at the beginning (left) and at the end of infiltration (right).

ples. The first value is obtained by averaging θ over (x,z) inside the sample area. The second value is obtained by subtracting the weight of the dry sample from the weight of the sample during the measurements and dividing it by the internal sample volume. The values of θ are plotted in Fig. 2. The two evaluations of θ match well in absolute values and temporal evolution. The tendency of the gravimetrically obtained θ to decrease a bit less than the imaged one is probably explained by the translocation of water from the soil into the aboveground biomass. This shows the advantage of the imaging technique to estimate changes in soil water content. Overall, the comparison of the two evaluations validates the QNI corrections and the quantitative estimation of θ by image analysis.

The water loss by evapotranspiration during day light was approximately equal to 5 g d^{-1} for the lupin and 2.8 g d^{-1} for the maize. During night light, the values decreased to 3 g d^{-1} for the lupin and to 1.68 g d^{-1} for the maize. The reference sample without plant showed an evaporation rate of 0.48 g d^{-1}.

3.2 Water distribution during infiltration

Figure 3 shows the water distribution at the beginning (t = 0) and at the end (t = 20 min) of the infiltration. The first line shows the sample with the maize and the second line the sample with the lupin. The top layers of both samples were initially much drier than the lower parts. This was caused by the fact that the upper layer was filled after planting, resulting in a less consolidated sand packing and thus in a lower water-holding capacity. Interestingly at the beginning of infiltration, more water was stored in the soil region near the roots. As water infiltrated, these regions continued to be wetter than the soil region far from the root. Gradients caused by root water uptake did not appear in the radiograms of the planted samples. This can be explained by the high soil conductivity and to the low transpiration rate.

In the sample with maize (Fig. 3, top line), dark spots are visible in the lower part of the container. The reason of these spots became obvious after opening the samples at the end of the measurements. The spots were precipitates on the container walls. Analysis with X-ray photoelectronic spectroscopy indicated that they were composed mostly by Al_2O_3 and SiO_2. Presence of Al indicates that corrosion of the container took place at these locations. The dark spots were visible from the beginning of the measurements and, as evaluated by differentiating two pictures taken at different times, the spots did not increase during time (data not shown). They probably formed during the previous preliminary tests with the same container. However, since the roots mostly did not

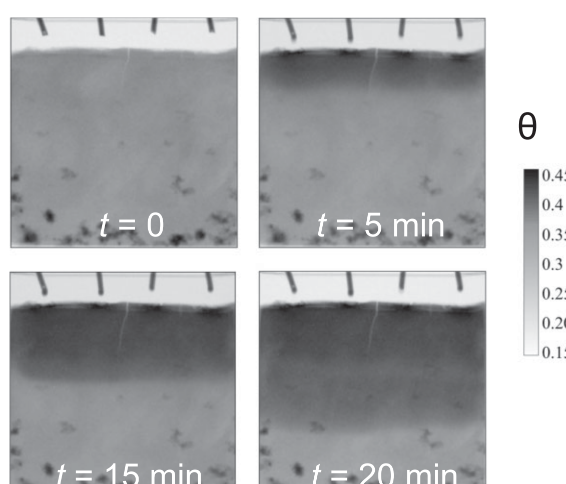

Figure 4: Time-series radiography of water infiltration into the unplanted sample. The small crack in the toplayer was caused by the suction induced by infiltration.

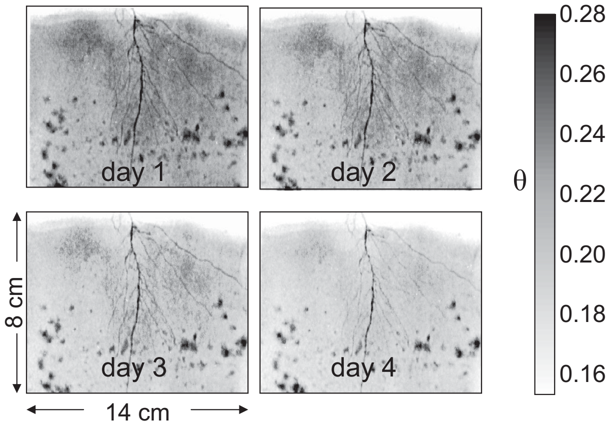

Figure 5: Radiographs of the sample with maize after infiltration. The radiographs were taken at 24:00 (midnight) for 4 d after irrigation.

reach the areas with the precipitates, the presence of these spots did not affect the water distribution near roots and our evaluation.

The unplanted sample showed a uniform infiltration front. This suggests that the nonuniform water infiltration in the planted sample was related to roots.

3.3 Water distribution during root water uptake

Figure 5 shows the radiographs of the sample with maize at midnight over 4 d after irrigation. During root uptake, we observed the same behavior as during infiltration: The soil near the roots stored more water than the soil far from the roots. Again, negative gradients of water content towards the roots were not visible, probably because of the high soil conductivity and the low uptake rate. The region next to the tip and to the apical part of the root dried first (day 2). A narrow region of water depletion developed along the main root where laterals were present (day 3). Overall, the portion of soil containing roots showed continuously the highest water contents, also during uptake.

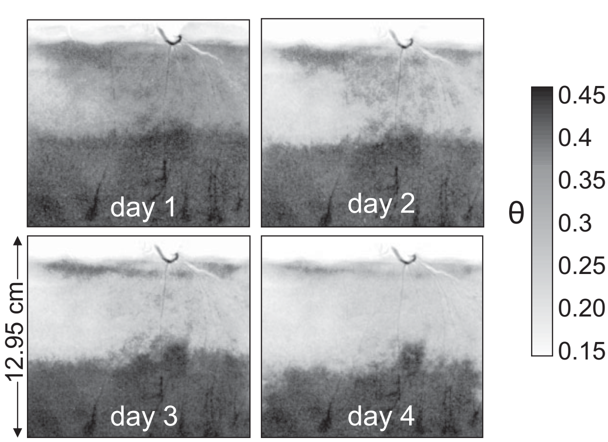

Figure 6: Radiographs of the sample with lupin after infiltration. The radiographs were taken at 24:00 (midnight) for 4 d after irrigation. A nonuniform drainage front developed during drying. High water content appeared at the root tips.

A similar behavior occurred for the sample with lupin (Fig. 6): More water was stored in the root region during all the observation period. Additionally, local regions of high water content were visible at the root tips. These dark regions were not caused by precipitates, but really indicate high content of water, or at least of hydrogen. They could be caused by mucilage exuded by the root tips, which is known to have an extremely high water-holding capacity, retaining up to 99% of water (*Young* et al., 1995; *Read* et al., 1997).

Figures 5 and 6 show only the radiographs taken at midnight. However, radiographs in the other times of the day showed the similar behavior. We did not observe a significant difference in the spatial distribution of soil moisture between day and night. This was probably due to the low transpiration rate of the samples and to the high soil conductivity.

3.4 Tomography of the water distribution at the root–soil interface

The rectangular slabs were tomographed in order to add the 3-D information of the root architecture to the time-series images obtained in a 2-D fashion by transmission. This combined approach has not been reported thus far in literature. The tomograms yield the 3-D gray-value distribution within the samples. Despite the rectangular shape of the samples, neutrons crossed the sample also through the longest side and the tomograms were reconstructed properly. Two vertical sections of the tomograms of the sample with lupin and with maize are shown in Fig. 7 and 8, respectively. Roots and heterogeneous water-content distributions were visible.

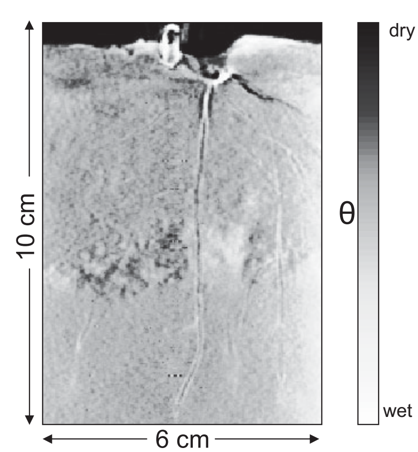

Figure 7: Vertical section of the tomogram of the sample with lupin. The colormap indicates the water content in the sample. The increase in darkness at the lateral edges of the sample is probably due to neutron scattering. Regions of water depletion are visible next to the more proximal part of the roots. Wet regions are visible at the more distal parts of the roots. The tomography was performed 79 h after infiltration. The average water content was 0. 29 $m^3\ m^{-3}$.

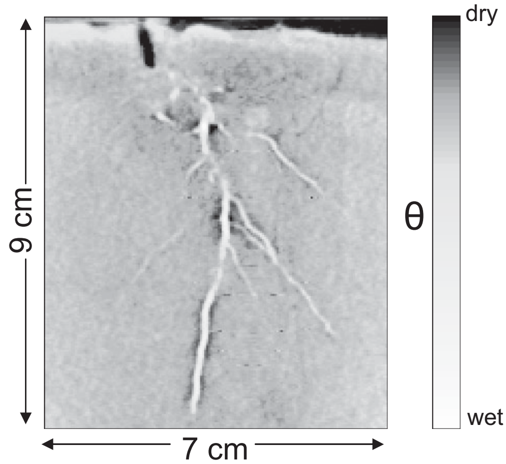

Figure 8: Vertical section of the tomogram of the sample with maize. The increase in darkness at the lateral edges of the sample is due to neutron scattering. Regions of water depletion are visible along the main root where laterals are present. The tomography was performed 22 h after infiltration. The average water content was 0.22 $m^3\ m^{-3}$.

The gray values in the tomograms are proportional to the neutron-attenuation coefficients of the materials. Since water has a neutron-attenuation coefficient much higher than the other soil components, the gray values are mainly related to the presence of water. Assuming a uniform soil density, the gradients in gray values are related to gradients in water content. In the images, dark means wet and white means dry. For our purposes, the tomograms are studied in order to visualize local variations in water content, rather than to obtain absolute values of water content.

In the tomograms, heterogeneous water-content distributions at the soil–root interface became clearly visible. In the sample with lupin, we observed a darker, wetter region adjacent to the apical part of the tap root with a uniform width of ≈ 2 mm. The region became darker, wetter at the tip of the root. In the upper region of the sample, we observed regions of water depletion around the more proximal parts of the roots, where laterals branched. This may indicate regions of root water uptake. This hypothesis is corroborated by the fact the root segments where root branches started are known to have high root hydraulic conductivity, therefore favoring the local water uptake. However, the observed dry regions could also be caused by a local, low water-holding capacity of the soil.

Regions of water depletion next to roots were visible especially in the sample with maize. In Fig. 8, at least three regions are evident where the rhizosphere was drier than the surrounding soil. Two regions of water depletion occurred along the main root where lateral branches started, and one region is present in the apical part of the main root. Interestingly, the root segments next to these dry areas were darker and wetter than the other root sections. As in the sample with lupin, water depletion appeared next to the root segments that are supposed to have a high radial conductivity. The spots visible at the bottom of the sample were due to the precipitates at the container walls.

The increase in darkness at the lateral edges of both tomograms was due to neutron scattering, which was not accounted for in the tomography reconstruction. However, this effect does not invalidate our analysis of the water distribution at the root–soil interface.

4 Discussion

This study shows the high capabilities of neutron radiography and tomography to image water distribution in the root zone with high contrast and spatial resolution. Radiography is particularly suited to investigate dynamic processes like water infiltration. Water-content distributions in two dimensions have been calculated from neutron radiography after correction for neutron scattering. Quantification of water content was confirmed by weighing the water loss in the sample. From the radiograms, we also imaged the root distribution. The radiograms showed that the root zones of both lupin and maize were significantly wetter than in the bulk soil during irrigation and in the subsequent drying period. Radiography of water infiltration in the unplanted sample showed a uniform wetting front, demonstrating that the heterogeneous water infiltration in the samples with plants was related to roots. Therefore, the fact that water preferentially wetted the root zone suggests that the hydraulic properties of the root zone were different from that of the bulk soil. Our hypothesis is that root activity increased the water-holding capacity of the root zone. This could be caused by root growth, which leads to soil compaction and pore-size decrease (*Bruand* et al., 1996), or by mucilage exuded by roots (*Young*, 1995; *Read* et al., 1997). Further studies are needed to investigate the mechanisms of how roots affect the soil hydraulic properties.

The radiograms did not show any water depletion around roots. Actually, we observed the same decrease in water content over time in the root zone and in the bulk soil (Fig. 4). This was probably caused by the low transpiration rate and the high soil conductivity.

Three-dimensional information was obtained by means of neutron tomography. Despite the rectangular shape of the sample and its large extension, the tomogram showed a good quality. The tomograms were particularly successful in showing the heterogeneous water distribution at the root–soil interface. Such a local imaging of moisture content in the rhizosphere (1–2 mm next to roots) was not possible *via* radiography. This is because radiography gives the average water distribution along the sample thickness (1.5 cm), therefore averaging the rhizosphere properties. The tomograms showed revealing heterogeneous water-content profiles along the roots. Dark regions were visible at the tips of the lupin. These dark regions indicate local higher water content, caused probably by an increased water-holding capacity of these portions of soils. The increase in the water-holding capacity is probably due to mucilage exuded by roots (*Young*, 1995; *Read* et al., 1997). Mucilage is mostly composed of polymeric substances, and it has a high water–holding capacity, explaining the dark, wet areas in the tomograms.

Regions of water depletion were visible along the main root of the maize, specifically where laterals branch, and at the apical part. It is known that where roots branch, the root conductivity is high, and this should increase the local uptake of water. The same applies to the apical part of the root, which is supposed to have a high conductivity. The observed water depletions could then indicate local water uptake by roots. A similar pattern of water depletion next to roots was observed in *Oswald* et al. (2009) and was interpreted as a quick uptake of water. However, there is no direct evidence that a gradient in water content indicates a flux. It may also indicate a region with varying hydraulic properties. Independent measurements of fluxes from soil to roots would be of great value for interpreting data on water-content distributions.

To summarize, this study shows the effect of plant roots on soil water distribution. By means of radiography, we observed that water preferentially infiltrated into the root zone and subsequently more water was retained there. This probably indicates that the root zone has distinct properties than the bulk soil. By means of neutron tomography, we imaged moisture gradients in the rhizosphere. High water contents next to roots were visible along the more distal parts of the lupin. Similar results have been reported (*Nakanishi* et al., 2005; *Tumlinson* et al., 2008). Local regions of water depletion have been observed along the main root of maize where root branched. These findings indicate that the moisture distribution in the rhizosphere varies significantly along the root system. This is probably caused by varying properties of rhizosphere and root and may play a crucial role on how water enters to roots. The rhizosphere properties and their affect on the location of root water uptake deserve further studies. With respect to this point, it would be of fundamental importance to support information on water distribution in the root zone with measurements of fluxes from soil to roots, as well as with measurements of root conductivity.

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